

Q1. Create a CSV file containing product names and prices. Read the data using Pandas. Use NumPy to calculate the average price. Write a function to identify products priced above the average and display them.

Concepts Used:

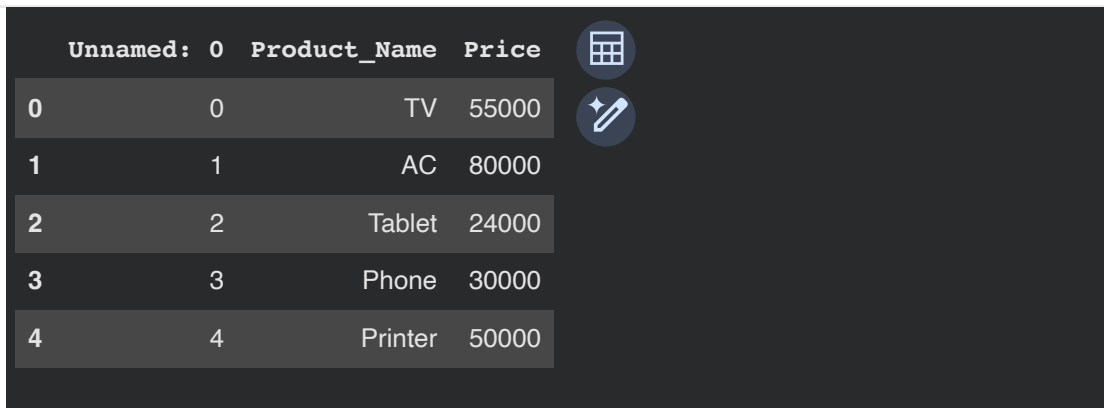
- Pandas DataFrame
- NumPy
- Functions
- Control Statements

```
import pandas as pd
import numpy as np
```

```
data = {
    "Product_Name": ["TV", "AC", "Tablet", "Phone", "Printer"],
    "Price": [55000, 80000, 24000, 30000, 50000 ]
}
```

```
df = pd.DataFrame(data)
df.to_csv("Products_detail.csv")

df = pd.read_csv("Products_detail.csv")
df
```

A screenshot of a Jupyter Notebook showing a DataFrame in table view. The table has five columns: 'Unnamed: 0', 'Product_Name', and 'Price'. There are five rows of data. To the right of the table are two icons: a grid icon and a pencil icon.

Unnamed: 0	Product_Name	Price
0	TV	55000
1	AC	80000
2	Tablet	24000
3	Phone	30000
4	Printer	50000

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
Average_price = np.average(df["Price"])
print("Average Price:", Average_price)
```

```
Average Price: 47800.0
```

```
def above_average(df, average_price):
    print("Product Above Average Price: ")
    for i in range(len(df)):
        if df["Price"][i] > average_price:
            print(df["Product_Name"][i], "-", df["Price"][i])

above_average(df, Average_price)
```

```
Product Above Average Price:
TV - 55000
AC - 80000
Printer - 50000
```

Q2. Read daily temperature data from a CSV file using Pandas. Use NumPy to calculate the highest, lowest, and average temperatures. Write functions to classify each day as:

- Cold (<20°C)
- Moderate (20–30°C)
- Hot (>30°C)

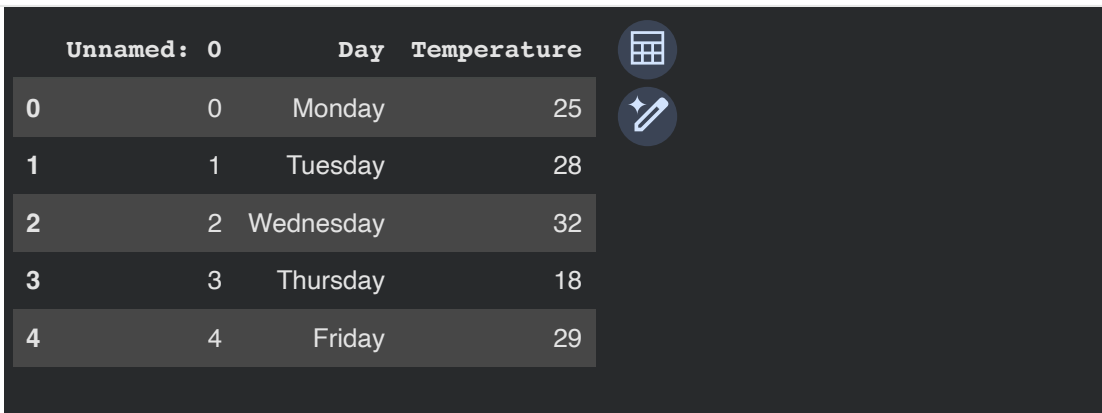
Display the category for each day. Concepts Used:

- Pandas
- NumPy
- Functions
- if-elif-else

```
import pandas as pd
import numpy as np
```

```
data = {  
    "Day": ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday"],  
    "Temperature": [25, 28, 32, 18, 29]  
}
```

```
df = pd.DataFrame(data)  
df.to_csv("Temperature_data.csv")  
  
df = pd.read_csv("Temperature_data.csv")  
df
```



The image shows a Jupyter Notebook table view of the DataFrame. It has a dark background with a light gray table. The table has four columns: 'Unnamed: 0', 'Day', 'Temperature', and an icon column. The rows are indexed from 0 to 4. To the right of the table are two icons: a grid icon and a pencil icon.

	Unnamed: 0	Day	Temperature	
0	0	Monday	25	
1	1	Tuesday	28	
2	2	Wednesday	32	
3	3	Thursday	18	
4	4	Friday	29	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
highest = np.max(df["Temperature"])  
print("Highest Temperature:", highest)  
lowest = np.min(df["Temperature"])  
print("Lowest Temperature:", lowest)  
average = np.mean(df["Temperature"])  
print("Average Temperature:", average)
```

```
Highest Temperature: 32  
Lowest Temperature: 18  
Average Temperature: 26.4
```

```
def classify_temp(df):  
  
    for i in range(len(df)):  
        temp = df['Temperature'][i]  
  
        if temp < 20:  
            category = "Cold"  
        elif temp <= 30:  
            category = "Moderate"  
        else:  
            category = "Hot"  
  
        print(df['Day'][i], "-", temp, "-", category)  
    classify_temp(df)
```

Monday - 25 - Moderate
Tuesday - 28 - Moderate
Wednesday - 32 - Hot
Thursday - 18 - Cold
Friday - 29 - Moderate